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Thermodynamics :- It can be defined as the science of heat energy and its transformations.

Thermo means heat energy

dynamics means transformations

It is broadly defined as the transformation of heat energy into mechanical and further electrical energy.

Bio Chemical Thermodynamics :-

The quantitative study of heat energy and its transductions within the living cells (or) Bio-chemical reactions.

All the organisms need of energy for their survival. That's why all the organisms capture, store, transduce and use of energy for their daily activities

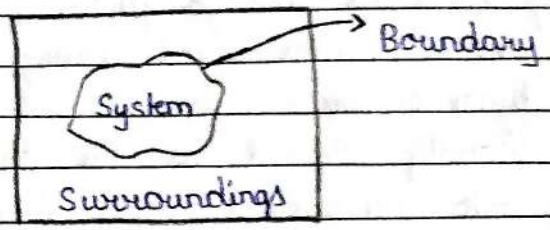
The solar energy is the main source of energy and is being captured by the green plants, certain bacteria, fungi, alage through photosynthesis about 2×10^{23} J/year which is 4% of solar energy.

Photosynthesis :-

Solar energy → Chemical Energy → Biological Macro molecules

$$6CO_2 + 6H_2O \xrightarrow[\text{Solar Energy}]{H_2O} C_6H_{12}O_6 + 6O_2$$

System :-



universe.

A substance (or) group of substances (or) a molecule (or) group of molecule in which we have self interest to study. (or) the part we study. is called System.

Example :- 1. Biological cells (or) group of cells
2. A Reactor (or) series of reactors

Surroundings :-

It is the part of universe and outside the system and separated by boundary is called Surroundings.

Boundary :-

It is an imaginary or real surface that separate system and surroundings.

Types of Systems :-

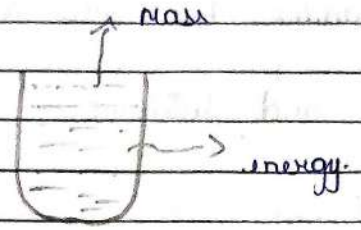
Thermodynamic systems are classified into open system, closed system and isolated system based on heat and mass transfer b/w system and surroundings

1. Open System :-

Mass transfer and heat transfer b/w system and surrounding under open conditions.

Ex :- An open cup of hot coffee.

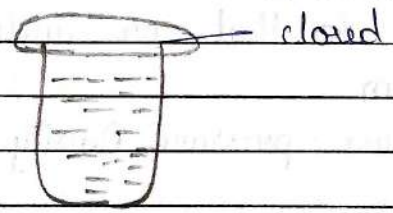
2. An open reactor



2. Closed System :-

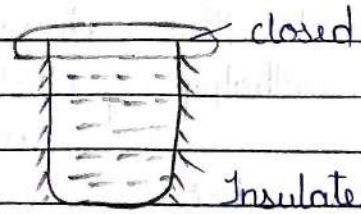
There is only heat transfer b/w system to surroundings

Ex :- 1. A closed cup of hot coffee
2. A closed reactor



3. Isolated System :-

There is no heat and mass b/w system to surroundings



Ex :- 1. An insulated and closed cup of hot coffee

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Properties of System

In Thermodynamics the properties of system are divided into

Extensive and Intensive

Extensive Properties

The properties highly depending on size or mass of the system

Ex:- Volume, Mass, Enthalpy, Entropy, P.E, K.E, light, Internal energy etc

Intensive Properties

The properties that are independent on size or mass of System

Ex- Temperature, pressure, Density

⇒ Some Intensive properties are derived by the ratio of any two extensive properties

Ex:- Density (ρ) = $\frac{\text{Mass } (m)}{\text{volume } (v)}$

⇒ Specific Intensive properties are derived by specifying the mass to the extensive property

Ex:- Specific enthalpy = $\frac{\text{enthalpy}}{\text{mass}}$

Specific K.E = $\frac{K.E}{\text{mass}}$

Specific heat = $\frac{\text{Heat}}{\text{mass}}$

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Energy Forms :-

Energy :-

It is the quantity that can be stored (or) exchanged b/w system and surroundings in the form of heat energy, mechanical energy, work, electrical, light energy and so on...

It is classified into :-

① Macroscopic forms

② Microscopic forms

① Macroscopic forms of energy :-

These energy forms are related to position, motion and pressure in the gravitational field

Ex :- P.E, K.E, Flow energy.

⇒ potential Energy :- Due to position of object in a gravitational field posses P.E

Mathematically $\boxed{P.E = mgh}$

⇒ kinetic Energy :- Due to motion of an object with certain velocity posses K.E

$$\boxed{K.E = \frac{1}{2} mv^2}$$

⇒ Flow Energy :- Due to fluid flow with certain pressure and volume posses F.E

$$\boxed{F.E = P.V}$$

② Microscopic forms of energy :-

These forms of energy are not physically visible and observable.

These are related to molecular structures, interaction

Ex :- Sensible, latent, chemical, ~~auto~~ atomic energies

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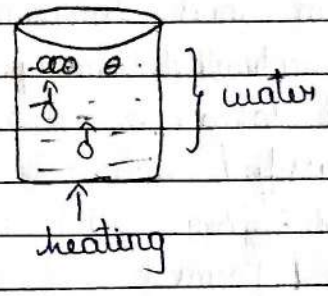
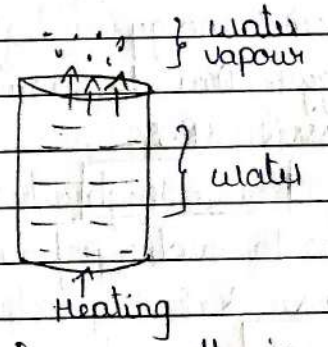
⇒ Sensible Energy :- The sum of vibrational, rotational and translational of molecules at high temperature is sensible heat.



⇒ Latent Energy :-

The amount of heat energy transfered during phase change



⇒ Chemical Energy It is associated with bonds of atoms in a molecule called as Bond energy

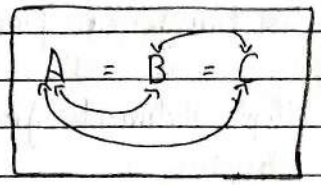
Ex :- Glucose $C_6H_{12}O_6$

⇒ Atomic Energy :- It is a anamous energy associated with nucleus of the atoms (or) Radio nuclides (U, Th, Ra ....)

Zero Law of Thermodynamics :-

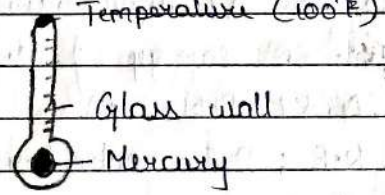
It is also known as law of Thermal Equilibrium

It states that if two systems are thermal equilibrium with third system then all three are thermal equilibrium with each other.



A is thermal equilibrium B and B thermal equilibrium with C then A is thermal equilibrium C without any direct contact

Ex:- Mercury Thermometer



When mercury thermometer is placed on human body then human body and glass wall attains thermal equilibrium due to conduction. Then after glass wall and mercury attains equilibrium through conduction. Finally the raising mercury shows temperature of human body that indicates mercury attains equilibrium with human body without direct contact.

Applications of Zeroth law of Thermodynamics

1. Design and development of clinical thermometer
2. Diagnosis Thermometers (mercury)
3. Digital Thermometers
4. Infra red Thermometers

s. pyro metru

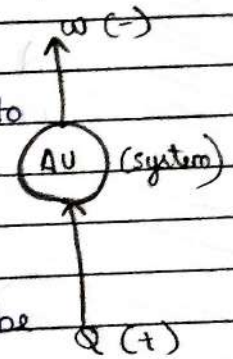
First law of Thermodynamics

It is also known as law of conservation of energy

It states that energy neither be created nor be destroyed but it can be transferred from one form to other forms.

Ex:- Let's apply this statement for non-flow process

As per first law of thermodynamics the change in energy of system must be equal to change in energy of surroundings



Δ(system) = Δ(surroundings)

The change in energy of system can be described as $\boxed{\Delta(K.E + P.E + T.E)}$

K.E = molecular K.E ; P.E = molecular P.E ; T.E = molecular Internal energy.

The change in energy of surroundings can be described in terms of amount of heat added and amount of work done by system

$\boxed{Q - W}$

Equating above two we can get

$$\boxed{\Delta(K.E + P.E + T.E) = Q - W}$$

For fluid non-flow process, K.E, P.E = 0

Therefore, $\boxed{\Delta(T.E) = Q - W}$

$$\boxed{\Delta U = Q - W \quad \text{—— (1)}}$$

ΔU - change in T.E

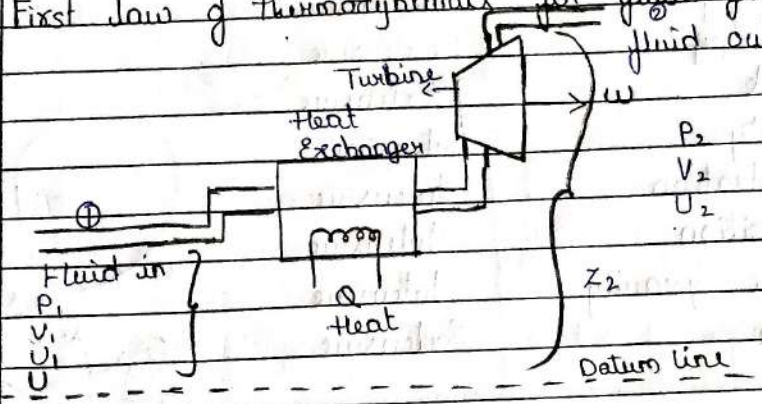
for differential form

$$\boxed{du = dQ - dw} \quad — ②$$

$$\boxed{\therefore dQ = du + dw} \quad — ③$$

From above eq ①, ②, ③ are called as mathematical statements of first law of thermodynamics

First law of thermodynamics for fluid flow process :-



As for law conservation of energy the total energy at section 1 (fluid inlet) = Total energy at S2 (fluid outlet).

When a fluid is flowing with a pressure (P), volume (V), velocity (u), internal energy (Q), flow energy (PV).

Additionally heat energy (Q) is added to the fluid in heat exchanger (HE) and work is released (W) from the turbine along with fluid outlet.

So that the total energy of the system at S1 is equal to the sum of macroscopic & microscopic forms of energy. i.e sum of K.E + P.E + I.E + F.E + Q

therefore:

$$\boxed{\frac{1}{2} m u_1^2 + (mg z_1) + U_1 + P_1 V_1 + Q} \quad — ①$$

Similarly the total energy of system at S_2

$$= K.E + P.E + I.E + F.E + W$$

$$= \boxed{\frac{1}{2} m u_2^2 + mg z_2 + U_2 + P_2 V_2 + W} \quad — ②$$

At steady conditions

Total energy S_1 = Total energy S_2

$$\frac{1}{2}mu_1^2 + mgz_1 + U_1 + Q + P_1V_1 = \frac{1}{2}mu_2^2 + mgz_2 + U_2 + P_2V_2 + W$$

(if m = 1 unit)

$$\frac{1}{2}u_1^2 + gz_1 + U_1 + P_1V_1 + Q = \frac{1}{2}u_2^2 + gz_2 + U_2 + P_2V_2 + W$$

$$Q - W = \frac{1}{2}(u_2^2 - u_1^2) + g(z_2 - z_1) + (U_2 - U_1) + (P_2V_2 - P_1V_1)$$

$$Q - W = \frac{\Delta u^2}{2} + g\Delta z + \Delta U + \Delta PV \quad \text{— (3)}$$

$$Q - W = \frac{\Delta u^2}{2} + g\Delta z + \Delta H \quad \text{— (4)}$$ (Enthalpy definition)

$$Q - W = \Delta H \quad (\text{or}) \quad \Delta H = W - Q \quad \text{— (5)}$$

$$dH = dQ - dW \quad \text{— (6)}$$

$$dQ = dH - dW \quad \text{— (7)}$$

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Enthalpy (H)

The total heat content of thermodynamic system is called as enthalpy

Mathematically it can be measured from following equation

$$H = U + PV$$

The change in enthalpy, $\Delta H = \Delta U + P\Delta V$

ΔU = change in I.E (J)

P = constant pressure (pascal)

ΔV = change in volume (m^3)

$Pa \times m^3 = J$

$\frac{N}{m^2} \times m^3 = J$

The change in enthalpy can be used to characterise chemical and Bio chemical reactions.

if $\Delta H < 0$ (Exergonic reaction)
Heat is liberated

if $\Delta H > 0$ (Endergonic reaction)

if $\Delta H = 0$ (Equilibrium reaction)

Enthalpy of products

$\Delta H = H_P - H_R$

Enthalpy of reactants

Comparision of fluid flow and Non-fluid flow w.r.t first law :-

Non-flow (closed)	Fluid flow (open)
$\Delta U = Q - W$	$\Delta H = Q - W$
$dU = dQ - dW$	$dH = dQ - dW$
$dQ = dU + dW$	$dQ = dH + dW$

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Ideal Gas :-

The gases do not have any intra or internal molecular or attractive forces

Eg :- Nobel gases

The behaviour of ideal gases can be studied through ideal gas such as :-

Boyle's law :-

$V \propto \frac{1}{P}$ (At const 'T')

Charle's law :-

$V \propto T$ (const P)

Avogadro's law :-

$V \propto n$ (const T, P)

Combining these three law we get

$PV \propto nT$

$PV = nRT$

Non-ideal gas :-

The PVT behaviour of non-ideal gas can be studied using highly complex equation.

Vanderwaal's eqn, Redlich-Kwong eqn, Redlich-kwong Soave eqn, Virial eqn.

Vanderwaal's equation

$$P + \frac{a}{q^2}(V-b) = RT$$

This is used to study behaviour of non-ideal gas like H_2, CO_2 ...

Redlich-Kwong equation

$$P = \left(\frac{RT}{V-b}\right) - \left(\frac{a}{\sqrt{T}\ V(V+b)}\right)$$

Modified form of vanderwaal's equation. More accurate than Vanderwaal eqn. It is widely used process engineering calculation.

Redlich - Kwong - Soave equation

$$p = \left(\frac{RT}{V-b}\right) - \frac{a\alpha T}{V(V+b)}$$

α - acentric factor

Determines the degree of sperricity from non-sperricity

$\alpha = 0$ (Ideal gas) $\rightarrow$ sperricity > non-sperricity

if $\alpha > 0$ (non-ideal) $\rightarrow$ non-sperricity > sperricity

Virial Equation

$$Z = \frac{PV}{RT} = 1 + BP + Cp^2 + DP^3$$

$$= 1 + \frac{B'}{V} + \frac{C'}{V^2} + \frac{D'}{V^3}$$

B, C, D', B', C', D' :- Virial co-efficient

Z :- compressibility factor

$= 1$ (Ideal gas)

$\left.\begin{array}{l} > 1 \\ < 1 \end{array}\right\}$ (Non ideal)

$$Z = \frac{V_{real\ gas}}{V_{ideal\ gas}}$$

$\Rightarrow$ Virial eqn is used in ~~eqn~~ design of equipment of reactor, Bio-reactor, fermentor, Distillation coloumn.

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Tutorial

1. In a process 1000J of heat is absorbed by a system and 600J of work is done by the system. What is the change in internal energy for the process.

$W = 600J \quad Q = 1000J$

$\Delta U = Q - W$

$\Delta U = 1000J - 600J$

$\Delta U = 400J$

2. A system (reactor) consisting of some fluid is stirred in a tank. The rate of work done on the system by the stirrer is 5hp. The heat generated due to stirring is dissipated to the surroundings. If the heat transferred to the surroundings is 3600 kJ/h. Calculate the change in internal energy.

$W = 5hp$

$W = 5 \times 745.4 = 3727 J/s$ (1hp = 745.4 J/s)

$Q = 3600 \ kJ/h$

$Q = 3600 \times 1000 / 3600$

$= 1000 J/s$

$\Delta U = -Q + W$

$= -1000 + 3727$

$= 2727 J/s$

3. Suppose a woman does 500J of work and 9500J of heat transfer occurs into environment in the process
a. What is the decrease in her internal energy. assuming no change in temperature (or) consumption of food.
b. What is her efficiency

Sol b. $W/Q \times 100$

$= 500/9500 \times 100$

$= 5.26\%$

a. $W = 500J$

$Q = 9500J$

$\Delta U = ?$

$\Delta U = -Q - W$

$= -9500 - 500$

$= 10{,}000J$